Supplementary Online Content

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Figure S1: Literature Search Strategies

Search strategies were developed with the assistance of a librarian. The filter employed is a modified version of the Highly Sensitive Search Strategy (HSSS) filter by Cochrane (Cochrane handbook 5.1, chapter 6.4.11.1) [1], incorporating an element of the Royle and Waugh Brief RCT Search Strategy (BRSS) [2], and the P3 filter developed by Cooper *et al*.[3]

Search strategy for MEDLINE (Ovid Interface):

Line	Search	Hits
1	Budesonide/	4341
2	Formoterol Fumarate/	1640
3	1 and 2	506
4	Budesonide, Formoterol Fumarate Drug Combination/	146
5	((budesonide adj3 formoterol) or symbicort or duoresp or fobumix or vannair).mp.	758
6	or/3-5	872
7	asthma/ or asthma, aspirin-induced/ or asthma, exercise-induced/ or asthma, occupational/ or status asthmaticus/	125127
8	(asthma* or wheez*).mp.	179820
9	7 or 8	179820
10	6 and 9	639
11	limit 10 to (clinical trial, phase iii or clinical trial or controlled clinical trial or multicenter study or randomized controlled trial)	241
12	(controlled clinical trial or randomized controlled trial).pt.	583372
13	random*.ab.	1060065
14	placebo.ab.	203130
15	clinical trials as topic/ or randomized controlled trials as topic/	313999
16	clinical trial/ or exp randomized controlled trial/	800055
17	trial.ti.	208826
18	clinical trial, phase iii/	15919
19	("Phase 3" or "phase3" or "phase III" or P3 or "PIII").tw,kw.	64471
20	or/12-19	1772717
21	10 and 20	355
22	11 or 21	364

Search strategy for EMBASE (Ovid interface):

ID	Search term(s)	Hits
1	budesonide/	20239
2	formoterol/	5809
3	formoterol fumarate/	1024
4	1 and (2 or 3)	2410
5	budesonide plus formoterol/	2416
6	((budesonide and formoterol) or symbicort or duoresp or fobumix or vannair).mp.	4531
7	or/4-6	4531
8	asthma/ or allergic asthma/ or aspirin exacerbated respiratory disease/ or asthmatic state/ or exercise induced asthma/ or experimental asthma/ or extrinsic asthma/ or intrinsic asthma/ or mild intermittent asthma/ or mild persistent asthma/ or moderate persistent asthma/ or nocturnal asthma/ or occupational asthma/ or severe persistent asthma/	249875
9	(asthma* or wheez*).mp.	295789
10	8 or 9	296008
11	7 and 10	3205
12	limit 11 to (clinical trial or randomized controlled trial or controlled clinical trial or multicenter study or phase 3 clinical trial)	913
13	random*.ab.	1438759
14	placebo.ab.	288866
15	"clinical trial (topic)"/ or "randomized controlled trial (topic)"/	268080
16	clinical trial/ or exp randomized controlled trial/	1249073
17	trial.ti.	283879
18	phase 3 clinical trial/	44175
19	("Phase 3" or "phase3" or "phase III" or P3 or "PIII").tw,kw.	115870
20	or/13-19	2499949
21	11 and 20	1216
22	12 or 21	1241

Search strategy for the Cochrane Central Register of Controlled Trials:

ID	Search	Hits
1	MeSH descriptor: [Budesonide] this term only	1674
2	MeSH descriptor: [Formoterol Fumarate] this term only	925
3	#1 AND #2	319
4	MeSH descriptor: [Budesonide, Formoterol Fumarate Drug Combination] this term only	149
5	((budesonide adj3 formoterol) or symbicort or symbiocort or duoresp or fobumix or vannair):ti,ab,kw	399
6	#3 OR #4 OR #5	639
7	MeSH descriptor: [Asthma] explode all trees	11093
8	(asthma* or wheez*):ti,ab,kw	33977
9	#7 OR #8	33977
10	#3 AND #9	226

Search strategy for ClinicalTrials.gov:

Condition or disease: Asthma
Other terms: Product containing budesonide and formoterol
Study type: Interventional studies (clinical trials)
Study results: All studies
Results: 116

Terms	Search Results*	Entire Database**
Synonyms		
product containing budesonide and formoterol	104 studies	191 studies
symbicort	100 studies	183 studies
formoterol and budesonide	4 studies	6 studies
budesonide-formoterol	3 studies	8 studies
Budesonide and formoterol product	1 studies	1 studies
product containing budesonide	112 studies	539 studies
budesonide	112 studies	534 studies
Pulmicort	13 studies	106 studies
Budecort	--	3 studies
Butacort	--	1 studies
Eltair	--	1 studies
entocort	--	16 studies
Nasocort	--	1 studies
Preferid	--	1 studies
Rhinocort	--	15 studies
Uceris	--	2 studies
formoterol	114 studies	505 studies
foradil	5 studies	50 studies
oxis	5 studies	17 studies
arformoterol	--	18 studies
Brovana	--	19 studies
oxeze	--	2 studies
Perforomist	--	6 studies
budesonide	112 studies	539 studies
Pulmicort	13 studies	106 studies
Budecort	--	3 studies
Butacort	--	1 studies
Eltair	--	1 studies
entocort	--	16 studies
Nasocort	--	1 studies
Preferid	--	1 studies
Rhinocort	--	15 studies
Uceris	--	2 studies
containing	9 studies	20,902 studies
contains	4 studies	6,161 studies
contain	2 studies	2,829 studies
contained	2 studies	1,352 studies
Comprise	1 studies	1,015 studies
product	26 studies	18,640 studies
asthma	116 studies	3,644 studies
Asthmatic	27 studies	537 studies

Figure S2: Certainty of Evidence

Certainty of evidence for each outcome was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) handbook and GRADEpro software [4, 5].

Risk of bias: Risk of bias for each outcome was assessed using the Cochrane Risk of Bias assessment tool (Appendix B). The contributed weight of the study to the meta-analysis was also considered. The certainty of evidence was downgraded for outcomes with serious or very serious risk of bias.

Imprecision: Imprecision refers to the degree of certainty around the best estimate of the absolute effect. The number of events, width of confidence intervals (narrow <0.10, moderate 0.10 to 0.50, wide >0.50), whether the confidence intervals included the possibility of small or no effect, and estimated Optimal Information Size (set with alpha 0.05 and power 0.8), were all used to determine whether the certainty of evidence for each outcome should be downgraded.

Inconsistency: Inconsistency refers to the degree of uniformity in the direction and magnitude of effects(s) across all studies. I^2 was reviewed in line with the Cochrane criteria (Cochrane handbook 5.1, chapter 10.10.2) [1]:

0% to 40%: might not be important;
30% to 60%: may represent moderate heterogeneity;
50% to 90%: may represent substantial heterogeneity;
75% to 100%: considerable heterogeneity.

Outcome trends per study were examined on forest plots. Study designs were reviewed for cause of inconsistency. Outcomes were downgraded if there was evidence of serious or very serious inconsistency.

Indirectness: Indirectness relates to whether the evidence answers the research question posed. Population, intervention, control, direct comparison, and outcomes were assessed. Certainty of evidence was downgraded for serious or very serious indirectness.

Publication bias: Publication bias was assessed and certainty of evidence downgraded if there was evidence of serious or very serious reporting bias or publication bias.

Base on the above assessment, the authors ascribed one of the four certainties to each outcome:

High: The true effect is similar to the estimated effect.

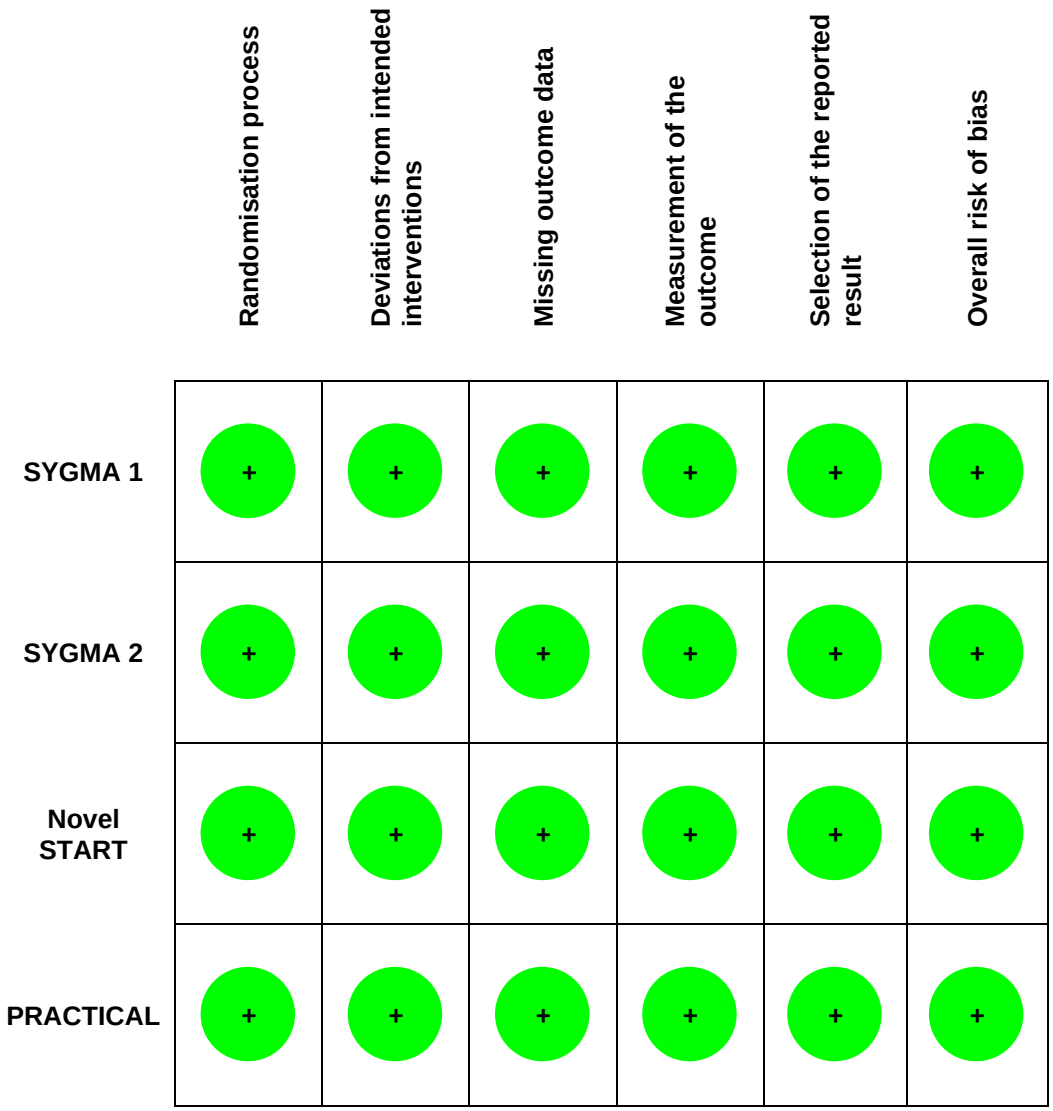
Moderate: The true effect is probably close to the estimated effect.

Low: The true effect might be markedly different from the estimated effect.

Very low: The true effect is probably markedly different from the estimated effect.

Figure S3: Risk of Bias Assessment

Risk of bias was assessed using the Cochrane Collaboration risk of bias tool for RCTs [6].



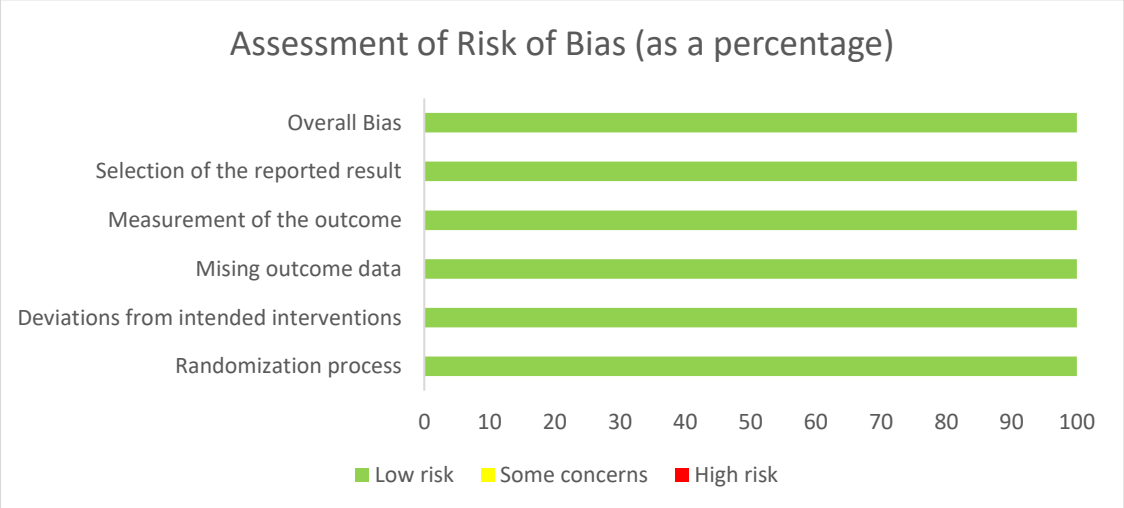


Table S1: Additional Study Characteristics

Characteristics	**SYGMA 1** [7, 8, 9]	**SYGMA 2** [7, 10, 11]	**Novel START** [12, 13, 14]	**PRACTICAL** [15, 16, 17]
Study registration	NCT02149199	NCT02224157	ACTRN12615000999538	ACTRN12616000377437
Participants, total - no.	3836	4176	648	885
Primary outcome, variable	Mean percentage of electronically recorded weeks with well-controlled asthma per patient	Annualized rate of severe exacerbations	Annualized rate of asthma exacerbations	Number of severe exacerbations per patient per year
Primary outcome, analysis	Superiority (with non-inferiority as secondary outcome)	Non-inferiority	Superiority	Superiority
Key inclusion criteria	1. Diagnosis of asthma according to GINA criteria, ≥6 month history; 2. Require GINA step 2 treatment; 3. FEV1: Pre-bronchodilator ≥60% predicted, post-bronchodilator ≥80% predicted, with reversibility; 4. Use of SABA ≥3 separate days in last week of run-in period.	1. Diagnosis of asthma according to GINA criteria, ≥6 month history; 2. Require GINA step 2 treatment; 3. FEV1: Pre-bronchodilator ≥60% predicted, post-bronchodilator ≥80% predicted, with reversibility; 4. Use of SABA ≥3 separate days in last week of run-in period.	1. Doctor diagnosis of asthma; 2. Use of SABA as sole therapy in preceding 3 months; 3. Use of SABA on ≥2 occasions, but ≤2 occasions per day, in the previous 4 weeks.	1. Doctor diagnosis of asthma; 2. Use of SABA with or without maintenance ICS (≤800µg budesonide equivalent) in the previous 12 weeks.
Key exclusion criteria	1. Smoking: ≥10 pack years; 2. History of life-threatening asthma; 3. Any significant disease or disorder which may put the patient at risk or influence the study results.	1. Smoking: ≥10 pack years; 2. History of life-threatening asthma; 3. Any significant disease or disorder which may put the patient at risk or influence the study results.	1. Use of non-SABA asthma medication in preceding 3 months; 2. Unstable asthma in 6 weeks prior to study (including use of oral steroids); 3. Diagnosis of other lung disease; 4. Smoking ≥20 pack years (or ≥10 with onset of respiratory symptoms after the age of 40 years).	1. Use of Non-ICS or SABA medication in preceding 3 months; 2. Unstable asthma in 6 weeks prior to study (including use of oral steroids); 3. Diagnosis of other lung disease; 4. Smoking ≥20 pack years (or ≥10 with onset of respiratory symptoms after the age of 40 years).

Characteristics	**SYGMA 1** [7, 8, 9]	**SYGMA 2** [7, 10, 11]	**Novel START** [12, 13, 14]	**PRACTICAL** [15, 16, 17]
Definition of severe exacerbation	Use of systemic corticosteroids for ≥3 days, or a hospitalization or ED visit because of asthma, requiring systemic corticosteroids	Use of systemic corticosteroids for ≥3 days, or a hospitalization or ED visit because of asthma, requiring systemic corticosteroids	Prescription of systemic corticosteroids for ≥3 days, or a hospitalization or ED visit because of asthma, requiring systemic corticosteroids	Use of systemic corticosteroids for ≥3 days, or a hospitalization or ED visit because of asthma, requiring systemic corticosteroids
Withdrawal of participants following a severe exacerbation	No	No	Yes	No
Run-in period, duration - weeks	2 to 4	2 to 4	N/A	N/A
Run-in period, medication	Terbutaline 500mcg as-needed only	Terbutaline 500mcg as-needed only	N/A	N/A
Treatment arm: as-needed ICS-formoterol	Budesonide-formoterol 200/6mcg (Symbicort Turbuhaler, AstraZeneca) one inhalation as-needed plus twice daily placebo	Budesonide-formoterol 200/6mcg (Symbicort Turbuhaler, AstraZeneca) one inhalation as-needed plus twice daily placebo	Budesonide-formoterol 200/6mcg (Symbicort Turbuhaler, AstraZeneca) one inhalation as-needed	Budesonide-formoterol 200/6mcg (Symbicort Turbuhaler, AstraZeneca) one inhalation as-needed
Treatment arm: maintenance ICS plus SABA reliever	Budesonide 200mcg (Pulmicort Turbuhaler, AstraZeneca) twice daily plus terbutaline 500mcg (Bricanyl Turbuhaler, AstraZeneca) as-needed	Budesonide 200mcg (Pulmicort Turbuhaler, AstraZeneca) twice daily plus terbutaline 500mcg (Bricanyl Turbuhaler, AstraZeneca) as-needed	Budesonide 200mcg (Pulmicort Turbuhaler, AstraZeneca) twice daily plus Albuterol 100mcg (Ventolin pMDI, GlaxoSmithKline) two inhalations as-needed	Budesonide 200mcg (Pulmicort Turbuhaler, AstraZeneca) twice daily plus terbutaline 250mcg (Bricanyl Turbuhaler, AstraZeneca) two inhalations as-needed
Treatment arm: SABA only	Terbutaline 500mcg (Bricanyl Turbuhaler) as-needed	NA	Albuterol 100mcg (Ventolin pMDI) two inhalations as-needed	NA
Funding source(s)	AstraZeneca	AstraZeneca	AstraZeneca and New Zealand Health Research Council	New Zealand Health Research Council

Abbreviations: ED, Emergency Department; ICS, Inhaled Corticosteroid; FEV_1, Forced Expiratory Volume over 1 second; SABA, Short-acting $beta_2$-agonist
All information derived from published trial protocols (including trial registries), manuscripts and supplementary material.

Table S2: Participant Baseline Demographics and Clinical Characteristics

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	220	225	437	448
Age, years								
Mean (range)	39.8 (12 to 80)	39.0 (12 to 85)	41.3 (12 to 83)	40.7 (12 to 83)	36 (18 to 74.4)	34.9 (18.3 to 74.1)	43.3 (18.3 to 74.7)	42.8 (18.1 to 75.8)
12 to <18 – no. (%)	161 (12.6)	173 (13.5)	205 (9.8)	206 (9.9)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
≥18 (%) – no. (%)	1116 (87.4)	1109 (86.5)	1884 (90.2)	1881 (90.1)	220 (100.0)	225 (100.0)	437 (100.0)	448 (100.0)
Female – no. (%)	777 (60.8)	797 (62.2)	1308 (62.6)	1289 (61.8)	122 (55.5)	129 (57.3)	244 (56)	241 (54)
Ethnicity – no. (%)								
White / European	721 (56.5)	728 (56.8)	1408 (67.4)	1406 (67.4)	168 (76.4)	171 (76.0)	342 (78.3)	357 (79.7)
Asian	340 (26.6)	347 (27.1)	399 (19.1)	403 (19.3)	14 (6.4)	25 (11.1)	29 (6.6)	34 (7.6)
Other, incl. Black, Māori or Pacific	216 (16.9)	207 (16.1)	282 (13.5)	278 (13.3)	38 (17.3)	29 (12.9)	66 (15.1)	57 (12.7)
Smoking status - no. (%)								
Current	34 (2.7)	30 (2.3)	53 (2.5)	54 (2.6)	18 (8.2)	22 (9.8)	39 (9)	24 (5)
Ex-Smoker	139 (10.9)	120 (9.4)	266 (12.7)	283 (13.6)	56 (25.5)	40 (17.8)	123 (28)	112 (25)
Never	1104 (86.5)	1132 (88.3)	1770 (84.7)	1750 (83.9)	146 (66.4)	163 (72.4)	275 (63)	312 (70)
Asthma medication – no. (%)								
SABA only	565 (44.2)	576 (44.9)	959 (45.9)	975 (46.7)	220 (100.0)	225 (100.0)	132 (30.2)	132 (29.5)
ICS or LTRA plus SABA	712 (55.8)	706 (55.1)	1130 (54.1)	1112 (53.3)	0 (0.0)	0 (0.0)	305 (69.8)	316 (70.5)
Severe exacerbations, prev. 12mths – no. (%)	257 (20.1)	241 (18.8)	459 (22)	460 (22)	12 (5.5)	17 (7.6)	53 (12)	52 (12)
ACQ-5 score – mean (SD)	1.6 (1.0)	1.6 (1.0)	1.5 (0.9)	1.5 (0.9)	1.1 (0.7)	1.1 (0.7)	1.1 (0.8)	1.2 (0.8)
FEV_1 (SD)*								
Mean, L	2.9 (0.8)	2.9 (0.8)	3.0 (0.8)	3.0 (0.8)	3.3 (0.9)	3.3 (0.9)	3.0 (0.9)	3.0 (0.9)

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
Percentage predicted value – mean	95.9 (14.0)	95.7 (13.4)	96.3 (13.8)	96.0 (13.5)	89.8 (14.1)	90.3 (13.6)	87.8 (16.4)	87.4 (16.3)
FeNO - ppb								
Mean (SD)	NA	NA	NA	NA	50.8 (46.1)	54.3 (44.0)	39.0 (33.9)	47.1 (44.8)
Median (IQR)	NA	NA	NA	NA	37.0 (18.0 to 66.0)	38.0 (20.0 to 76.0)	26.0 (15.0 to 51.0)	30.0 (18.0 to 62.5)
Range	NA	NA	NA	NA	3.0 to 300.0	5.0 to 200.0	5.0 to 222.0	5.0 to 300 .0

Abbreviations: ACQ-5, Asthma Control Questionnaire; FEV1, Forced Expiratory Volume over one second; LTRA, Leukotriene Receptor Antagonists; SD, standard deviation.

*SYGMA 1 and SYGMA 2 reported pre-bronchodilator FEV1 measurements; Novel-START and PRACTICAL reported on-treatment FEV1 measurements.

All data as published in study manuscripts and supplementary material.

Table S3: Severe Exacerbations

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	220	225	437	448
Severe exacerbations (n)	**(1277)**	**(1282)**	**(2089)**	**(2087)**	**(220)**	**(225)**	**(437)**	**(448)**
Total no.	77	89	217	221	9	21	48	68
Patients with ≥1	71	78	177	184	9	21	37	59
Time to first (HR)	0.90 (95% CI 0.65 to 1.24)		0.96 (95% CI 0.78 to 1.17)		0.41 (95% CI 0.19 to 0.90)		0.60 (95% CI 0.40 to 0.91)	
Annualised rate	0.07	0.09	0.11	0.12	0.05	0.12	0.12	0.17
Rate ratio	0.83 (95% CI 0.59 to 1.16)		0.97 (95% CI 0.78 to 1.20)		0.41 (95% CI 0.19 to 0.90)		0.69 (95% CI 0.48 to 1.00)	
ED visits and Hospitalisations (n)	**(1277)**	**(1282)**	**(2089)**	**(2087)**	**(220)**	**(225)**	**(437)**	**(448)**
ED visits (with SCS) – ≥1 visit	7	10	25	36	0	3	4	7
ED visits (with SCS) – total no.	8	10	26	40	0	3	5	7
Hospital admissions – ≥1 admission	6	8	17	17	0	0	0	2
Hospital admissions – total no.	6	8	20	17	0	0	0	2
ED or hospital admissions – ≥1*	12	16	39	48	0	3	4	8
ED or Hospital admissions – total no.*	13	16	44	52	0	3	5	9

Abbreviations: ED, Emergency Department (for asthma, requiring the use of systemic corticosteroids); HR, Hazard Ratio; SCS, systemic corticosteroids.
*Data for combined ED visits or hospitalisation were provided by the SYGMA study authors, on request.

Table S4: Serious Adverse Events

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	222	227	440	448
Patients with ≥1 SAE	38	37	66	73	11	6	25	15
Events – total no.	**49**	**47**	**80**	**87**	**16**	**7**	**28**	**18**
Cardiac disorders	2	4	4	5	0	0	6	1
Congenital, familial and genetic disorders	0	0	0	0	0	0	0	1
Ear and labyrinth disorders	0	0	0	1	0	0	0	0
Eye disorders	0	0	3	1	0	0	0	0
Endocrine disorders	0	2	0	0	0	0	0	0
Eye disorders	1	1	0	0	0	0	0	0
Gastrointestinal disorders	7	4	1	3	2	0	4	1
General disorders and administration site conditions	0	0	0	0	0	0	1	0
Hepatobiliary disorders	1	2	1	6	0	0	0	1
Immune system disorders	1	0	0	2	0	0	0	0
Infections and infestations	10	11	11	14	2	2	6	2
Injury, poisoning and procedural complications	4	3	14	13	1	0	2	5
Metabolism and nutrition disorders	0	0	0	1	4	0	0	0
Musculoskeletal and connective tissue disorders	1	1	4	4	0	1	1	1

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
Neoplasms benign, malignant and unspecified (including cysts and polyps)	6	1	4	4	1	0	3	1
Nervous system disorders	3	2	3	5	0	0	0	2
Product Issues	0	1	0	0	0	0	0	0
Psychiatric disorders	0	0	1	0	4	2	0	0
Renal and urinary disorders	1	2	1	1	0	0	1	0
Reproductive system and breast disorders	3	1	2	6	2	0	2	1
Respiratory, thoracic and mediastinal disorders	7	10	25	18	0	1	1	2
Skin and subcutaneous tissue disorders	1	1	2	0	0	0	0	0
Surgical and medical procedures	0	0	0	0	0	0	1	0
Vascular disorders	1	1	4	3	0	1	0	0

Abbreviations: SAE, Serious adverse events.
Terminology from the Medical Dictionary for Regulatory Activities (MedDRA). All data as published in study manuscripts, supplementary material, and trial registries.

Table S5: Deaths

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	222	227	440	448
Events – total no.	**0**	**2**	**1**	**1**	**1**	**1**	**0**	**0**
Brain Neoplasm	0	1	0	0	0	0	0	0
Upper gastrointestinal haemorrhage	0	1	0	0	0	0	0	0
Cardiorespiratory arrest	0	0	1	0	0	0	0	0
Acute asthma exacerbation	0	0	0	1*	0	0	0	0
Suicide	0	0	0	0	0	1	0	0
Road traffic incident	0	0	0	0	1	0	0	0

All data as published in study manuscripts and supplementary material. *Asthma-related death

Table S6: Inhaled Medication Use

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	220	225	437	448
Daily dose of IP ICS (n)	**(1277)**	**(1282)**	**(2089)**	**(2087)**	**(220)**	**(225)**	**(55)**	**(55)**
Mean - no. (SD)	92.9 (102.2)	314.9 (89.2)	103.5 (109.3)	250.6 (117.6)	107 (109)	222 (113)	176 (143)	302.5 (84.8)
Median (IQR / 95% CI)*	56.9 (50.0 to 64.6)	339.8 (332.1 to 346.7)	65.6 (60.8 to 71.7)	267.2 (256.5 to 272.0)	73.1 (30.7 to 146.3)	247 (132.0 to 314.0)	164.3 (74.0 to 251.7)	328.3 (245.8 to 364)
Range	0 to 721	0 to 701	0 to 753	0 to 743	0 to 790.2	0 to 402	6.7 to 682.5	26.8 to 458.1
Adherence – mean percentage dose taken	79	79	64	63	NA	56	NA	76
Daily beta$_2$-agonist-containing actuations (n) ¥	**(1276)**	**(1281)**	**(2089)**	**(2084)**	**(220)**	**(225)**	**(55)**	**(55)**
Mean (SD)	0.47 (0.51)	0.40 (0.56)	0.52 (0.55)	0.49 (0.70)	0.53 (0.54)	0.52 (1.03)	0.9 (0.7)	0.5 (0.6)
Median (IQR)	0.29 (0.07 to 0.72)	0.16 (0.04 to 0.52)	0.33 (0.09 to 0.79)	0.21 (0.05 to 0.65)	0.37 (0.15 to 0.73)	0.18 (0.06 to 0.46)	0.8 (0.4 to 1.3)	0.3 (0.1 to 0.6)
Min to max	0.0 to 3.6	0.0 to 3.7	0.0 to 3.8	0.0 to 7.2	0 to 3.95	0.0 to 8.7	0.0 to 3.4	0.0 to 2.7

Abbreviations: 95$ CI, 95% Confidence Interval; ICS, Inhaled Corticosteroid; IP, Investigational Product; IQR, Interquartile range; SD, standard deviation.
*95% Confidence Intervals reported in SYGMA 1 and SYGMA 2 publications; Interquartile Range reported in Novel START and PRACTICAL publications.
¥For analysis, formoterol 6mcg = salbutamol 200mcg = terbutaline 500mcg. Data for daily beta2-agonist-containing actuations from the SYGMA 1 and SYGMA 2 studies were provided by the study authors, on request.
All other data as reported in study manuscripts and supplementary material.

Table S7: Asthma Control and Lung Function

	SYGMA 1 [7, 8, 9]		**SYGMA 2** [7, 10, 11]		**Novel START** [12, 13, 14]		**PRACTICAL** [15, 16, 17]	
	Budesonide-Formoterol	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**	**Budesonide-Formoterol**	**Maintenance Budesonide**
No. in treatment arm	1277	1282	2089	2087	220	225	437	448
ACQ-5 - mean difference	0.149 (95% CI 0.101 to 0.198)		0.11 (95% CI 0.07 to 0.15)		0.14 (95% CI, 0.05 to 0.23)		0.06, 95% (CI -0.005 to 0.12)	
FEV1* - mean difference	-54.3mL (95% CI -78.8 to -29.8)		-32.6mL (95% CI -53.7 to -11.4)		0.004L (95% CI -0.03 to 0.04)		0.006L (-0.026 to 0.04)	

Abbreviations: ACQ-5, Asthma Control Questionnaire; FEV1, Forced Expiratory Volume over one second.
All data from published manuscripts and supplementary material. Published calculations used mixed linear models for continuous and repeated measures.

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